# LC Dynamics Simulator Technical Manual

## 1. Scope

This document describes the physical model, discretization, numerical solvers, optical calculation, validation, and limits of LC Dynamics Simulator for researchers and engineers. See the separate User Manual for GUI operation.

The spatial domain is one-dimensional through cell thickness $z$, while the director has three components. User-defined surface orientation, azimuth, twist, chiral pitch, and material constants make the model applicable beyond TN configurations, including STN, ECB, and VA ($\Delta\varepsilon < 0$), within these constraints:

- The field is along $z$ only; in-plane electrode fields (IPS, FFS) are not modeled.
- Both substrates share one surface polar angle $\theta$ with strong anchoring; hybrid alignment is not representable.
- The polarizer is along $x$ and the analyzer along $y$ (crossed, fixed).

Lateral nonuniformity, defects, flexoelectricity, ion transport, and thermal fluctuations are also outside the current model.

## 2. Coordinates and Initialization

The director is

$$\mathbf{n}(z,t) = (n_x, n_y, n_z), \qquad |\mathbf{n}| = 1.$$

The GUI's Surface tilt $\theta$ is the polar angle of the director measured from the substrate normal $z$ ($n_z = \cos\theta$), shared by both substrates: $\theta = 90^\circ$ is parallel to the substrate, $0^\circ$ perpendicular, and the default $88^\circ$ corresponds to about $2^\circ$ pretilt. The tilt shown in the GUI profile chart is measured from the substrate plane,

$$\vartheta = \arcsin n_z,$$

so $\vartheta = 90^\circ - \theta$ at the substrates. The azimuth is set by Bottom azimuth $\phi_0$ (measured from $x$, the polarizer) and Twist angle $\Delta\phi$.

With $N$ grid points (GUI Grid points), nodes $k = 0, \dots, N-1$ are used and the solver spacing is $h = d/(N-1)$. The potential is solved over the $N-1$ intervals from node $0$ ($V_{\text{com}}$) to node $N-1$ ($V_{\text{pix}}$). The director lives on nodes $k = 1, \dots, N-1$, with $k = 1$ and $k = N-1$ fixed at the substrates. The $z$ coordinates in the GUI and CSV place nodes $1, \dots, N-1$ on $0 \le z \le d$ with spacing $d/(N-2)$. This grid follows the C and Python versions. The initial estimate uses

$$\phi_k = \phi_0 + \Delta\phi\,\frac{k-1}{N-2},$$

$$n_x = \sin\theta\cos\phi_k, \quad n_y = \sin\theta\sin\phi_k, \quad n_z = \cos\theta.$$

Strong anchoring fixes the substrate directors. Interior nodes are relaxed at zero field before time integration.

## 3. Elastic and Dielectric Model

The Frank-Oseen elastic energy is

$$f_{\text{el}} = \frac{K_{11}}{2}(\nabla \cdot \mathbf{n})^2 + \frac{K_{22}}{2}(\mathbf{n} \cdot \nabla \times \mathbf{n} + q_0)^2 + \frac{K_{33}}{2}|\mathbf{n} \times \nabla \times \mathbf{n}|^2, \qquad q_0 = \frac{2\pi}{p}.$$

Setting Chiral pitch $p = 0$ in the GUI gives $q_0 = 0$ and disables the chiral term. Only $z$-derivatives remain. The dielectric tensor is

$$\varepsilon = \varepsilon_\perp \mathbf{I} + (\varepsilon_\parallel - \varepsilon_\perp)\mathbf{nn},$$

and the one-dimensional electrostatic equation is

$$\frac{d}{dz}\left(\varepsilon_{zz}\frac{dV}{dz}\right) = 0, \qquad \varepsilon_{zz} = \varepsilon_\perp + \Delta\varepsilon n_z^2.$$

With fixed electrode potentials, the discrete equation is tridiagonal. Accurate mode uses C-order iterative relaxation; Fast mode uses a Thomas direct solve.

The capacitance per unit area is the series combination of the node layers,

$$C = \left(\sum_{k=1}^{N-1} \frac{h}{\varepsilon_{zz,k}}\right)^{-1},$$

reported in $\mu\text{F}/\text{m}^2$.

## 4. Director Dynamics and Backflow

### 4.1 Kinematic Quantities

Leslie-Ericksen theory treats director **n** and velocity **v** as coupled continuum fields. The velocity gradient is split into the symmetric rate-of-strain tensor and antisymmetric vorticity tensor,

$$A_{ij} = \frac{1}{2}(\partial_j v_i + \partial_i v_j), \qquad \Omega_{ij} = \frac{1}{2}(\partial_j v_i - \partial_i v_j).$$

The director rotation relative to local rigid-body rotation is

$$N_i = \frac{Dn_i}{Dt} - \Omega_{ij} n_j, \qquad \frac{D}{Dt} = \frac{\partial}{\partial t} + v_j \partial_j.$$

The molecular field is the variational derivative of free energy,

$$h_i = -\frac{\delta F}{\delta n_i}.$$

It represents the generalized elastic and electric torque that drives the director toward lower free energy.

### 4.2 Director Torque Balance

With director inertia neglected, the torque balance in the convention used here is

$$h_i - \lambda n_i = \gamma_1 N_i + \gamma_2 A_{ij} n_j,$$

or, after removing the arbitrary component parallel to the director,

$$\mathbf{n} \times (\mathbf{h} - \gamma_1 \mathbf{N} - \gamma_2 \mathbf{A}\mathbf{n}) = \mathbf{0}.$$

The Lagrange multiplier $\lambda$ enforces $|\mathbf{n}| = 1$. The $\gamma_1 \mathbf{N}$ term is rotational viscous resistance; $\gamma_2 \mathbf{A}\mathbf{n}$ describes alignment or rotation by shear and extension; and $\mathbf{h}$ contains elastic and dielectric driving torque. The rotational combinations are

$$\gamma_1 = \alpha_3 - \alpha_2, \qquad \gamma_2 = \alpha_6 - \alpha_5.$$

Positive $\gamma_1$ gives rotational dissipation, while $\gamma_2/\gamma_1$ is related to flow-aligning or tumbling behavior in simple shear.

### 4.3 Momentum Balance and Leslie Stress

Incompressibility and momentum conservation are

$$\partial_i v_i = 0,$$

$$\rho \frac{Dv_i}{Dt} = -\partial_i p + \partial_j (\sigma^{\mathsf{L}}_{ij} + \sigma^{\mathsf{E}}_{ij}).$$

Here $p$ enforces incompressibility, $\sigma^{\mathsf{L}}$ is the Leslie viscous stress, and $\sigma^{\mathsf{E}}$ is the Ericksen elastic stress. The present implementation applies the electric field through the molecular field and does not add a separate Maxwell body force to the momentum equation.

Using the index convention of this manual,

$$\begin{aligned}\sigma^{\mathsf{L}}_{ij} = {} & \alpha_1 n_i n_j n_k n_l A_{kl} + \alpha_2 n_i N_j + \alpha_3 N_i n_j + \alpha_4 A_{ij} \\ & + \alpha_5 n_i n_k A_{kj} + \alpha_6 A_{ik} n_k n_j.\end{aligned}$$

| Coefficient | Principal physical role |
|---|---|
| $\alpha_1$ | Anisotropic extension or compression along the director |
| $\alpha_2, \alpha_3$ | Coupling between relative director rotation and shear stress |
| $\alpha_4$ | Isotropic-like shear-viscosity contribution |
| $\alpha_5, \alpha_6$ | Anisotropic shear response parallel and transverse to the director |

Thermodynamically consistent conventional nematics normally satisfy the Parodi relation

$$\alpha_6 - \alpha_5 = \alpha_2 + \alpha_3.$$

The app accepts all six coefficients independently and does not enforce the Parodi relation or dissipation in-

equalities. Researchers must verify coefficient sign convention, temperature, frequency, and thermodynamic consistency before use.

A representative form of the Ericksen stress for free-energy density $f$ is

$$\sigma^{\mathsf{E}}_{ij} = -\frac{\partial f}{\partial(\partial_j n_k)}\,\partial_i n_k.$$

Isotropic terms that can be absorbed into pressure differ among conventions, so published stress expressions can appear different while giving the same torque and force balance.

#### 4.4 One-dimensional Reduction

The implementation assumes

$$\mathbf{v} = (u_x(z,t), u_y(z,t), 0), \qquad \mathbf{n} = \mathbf{n}(z,t).$$

Incompressibility is then automatic, and the principal strain-rate components are

$$A_{xz} = A_{zx} = \frac{1}{2}\frac{\partial u_x}{\partial z}, \qquad A_{yz} = A_{zy} = \frac{1}{2}\frac{\partial u_y}{\partial z}.$$

No-slip boundaries impose $u_x = u_y = 0$ at the substrates. The code uses the C-compatible component expansion of these continuum equations. Each time step iterates dielectric response and potential, molecular field, the unit-length multiplier, all three director components, and $u_x, u_y$ to convergence.

#### 4.5 Meaning of Backflow

Director rotation generates Leslie stress, particularly through the $\alpha_2$ and $\alpha_3$ terms, and therefore drives flow. The resulting velocity gradient acts back on the director through **A** and $\Omega$. This bidirectional coupling is backflow and can produce nonmonotonic local orientation or optical response after a voltage transition.

## 5. Drive Functions

The voltage of step $i = 0, 1, \ldots$ is evaluated at the end of the step, $t = (i+1)\Delta t$. Step off applies $V_{\text{high}}$ while $i$ is at most Voltage-off step and $V_{\text{low}}$ afterward. Square wave uses period $P$, duty $D$, and phase. Waveform mode evaluates user rows $(t_i, V_i)$ with either sample-and-hold

$$V(t) = V_i \quad (t_i \le t < t_{i+1})$$

or linear interpolation

$$V(t) = V_i + (V_{i+1} - V_i)\frac{t - t_i}{t_{i+1} - t_i}.$$

$V_1$ applies before the first row and $V_N$ after the last. Equal-time rows retain table order, and the later row takes effect, which represents an instantaneous transition.

## 6. Numerical Method

Static equilibrium uses relaxed fixed-point iteration,

$$x \leftarrow x + a(x^* - x),$$

with a default relaxation factor of 0.5 and $10^{-6}$ director candidate tolerance. Dynamic integration uses a fixed user-supplied time step and nested convergence iterations.

Accurate mode retains the original voltage-relaxation order and tighter tolerances. Fast mode uses the direct voltage solve and looser director and auxiliary tolerances. Fast is intended for exploration; Accurate is recommended for reported results.

Research use should include grid convergence, time-step convergence, relaxation-factor sensitivity, and an Accurate/Fast comparison. Profile stride and live-update stride affect storage and rendering, not the physical time stepping.

## 7. Normal-Incidence Optics

Each layer is treated as a uniaxial retarder and accumulated with 2×2 Jones matrices. The azimuth is

$$\phi = \mathrm{atan2}(n_y, n_x),$$

the layer effective index is

$$n_{\mathrm{eff}} = \left( \frac{\cos^2 \vartheta}{n_e^2} + \frac{\sin^2 \vartheta}{n_o^2} \right)^{-1/2},$$

with $\vartheta = \arcsin n_z$ the tilt from the substrate plane, and phase retardation is

$$\delta = \frac{2\pi \Delta z}{\lambda} (n_{\mathrm{eff}} - n_o).$$

Crossed x-polarizer/y-analyzer transmission is $T = |E_y|^2$. Interface Fresnel loss, absorption, multiple reflection, scattering, and compensation films are excluded.

## 8. Oblique Incidence and Contrast

For viewing direction

$$\mathbf{k} = (\sin\Theta\cos\Phi, \sin\Theta\sin\Phi, \cos\Theta),$$

with polar angle $\Theta$ and azimuth $\Phi$ measured from $x$, the director and polarizer/analyzer axes are projected into the plane transverse to **k**. The extraordinary index is calculated from the angle between director and propagation direction, and optical path length is scaled by $1/\cos\Theta$.

$\Theta$ is the propagation angle inside the liquid-crystal layer; refraction at the air, glass, and liquid-crystal interfaces (Snell's law) is not modeled. To compare with an observation angle in air, $\Theta_{\mathrm{air}}$, convert it yourself, for example with $\sin\Theta_{\mathrm{air}} = \bar{n}\sin\Theta$ for an average index $\bar{n}$.

Contrast is

$$CR(\Theta, \Phi) = \frac{T_{\mathrm{bright}}}{\max(T_{\mathrm{dark}}, 10^{-9})}.$$

Report bright and dark transmittance as well as the ratio when the denominator approaches the numerical floor.

## 9. Software Architecture

The Swift package `LCSimulatorCore` isolates the numerical engine. `Director.swift` contains initialization, static equilibrium, and electrostatics; `DynamicStep.swift` contains the coupled step; `Simulation.swift` manages the time loop; `Optics.swift` and `OpticalMap.swift` implement optical calculations; `Waveform.swift`, `CSVExport.swift`, and `ParameterFile.swift` handle input/output.

The SwiftUI app runs calculations in the background. The Mac App Store build uses no JIT compilation.

## 10. Validation

Automated regression data were generated with the Python `python-final` tag. Tests cover Accurate/Step off, Fast/Square wave, time response, final profiles, Jones vectors, viewing-angle maps, contrast maps, waveform equivalence, CSV format, progress events, and cancellation.

Because Numba fastmath and Swift evaluate floating-point expressions in different orders, tests use relative tolerance $10^{-6}$ and absolute tolerance $10^{-8}$, rather than bit equality. Current reference differences are well below these limits.

## 11. Output Schemas

```
step,time_ms,voltage_V,capacitance_uF_per_m2,transmittance
polar_angle_deg,azimuth_deg,time_ms,transmittance
polar_angle_deg,azimuth_deg,bright_time_ms,dark_time_ms,contrast_ratio
```

Angular maps use one row per polar/azimuth pair. CSV uses UTF-8 with BOM, CRLF line endings, and 12 significant digits.

## 12. Model Limitations

- One spatial dimension through the cell thickness
- Strong anchoring with one surface polar angle shared by both substrates; isothermal, incompressible treatment
- Fixed crossed polarizer ($x$) and analyzer ($y$)
- Viewing-angle optics without interface refraction; polar angles are propagation angles inside the liquid crystal
- No lateral field, lateral nonuniformity, defects, ions, charge injection, conductivity anisotropy, or flexoelectricity
- Jones optics without scattering, absorption, Fresnel interfaces, or multiple reflections
- Temperature, frequency, and wavelength dependence must be represented through user inputs
- Numerical convergence does not establish physical applicability

## 13. References


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